Q.1. Implement the decision tree classifier using ID-3 algorithm for the Mushroom Dataset (agaricus-lepiota.data) provided within the assignment folder. I have also included a PDF file containing useful description of the dataset downloaded from the following link: https://archive.ics.uci.edu/ml/datasets/mushroom. Use 5-fold cross validation to report performance. Moreover, use appropriate measures to handle attributes with missing values.

Q.2. Implement a random forest classifier for the classification problem in Q.1. Use the square root of the total features (rounded up) to induce individual trees.

Q.3. Implement a decision tree regressor for the provided dataset. Use the mean squared error as the cost function to maximize the gain.

Q.4. Use locally weighted regression to find a non-parametric model to estimate the cost of house. Compare its performance with that of the decision tree regressor in Q.3.

Submission Notes:

1. The deadline for submission is 11:59 PM on Oct 22, 2026
2. Submit a soft copy along with the code through email: ali.ds@pu.edu.pk